

EE 419 Project 4 Design Document

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Overall System

Our overall system works with a forever loop that listens for connections. Before starting, the entire web root directory ('./contents') is searched for files. Each file is classified as protected or not protected, and the contents themselves are stored along with the date modified and the contents length. When a new TCP connection starts, a new thread is launched to handle all messages for that socket. When messages come in for that connection, a special handler takes the message sent over TCP and processes it. The first line is split up so that the path is extracted. The headers are then parsed for connection closed and range. First the server checks if the file exists. If not, a 404 is returned. If it does exist, then the permissions are checked. If it is protected, a 403 is returned. Then the file size is checked. If it is more than 5MB, then a 206 is returned with the range (0-5MB). If the range was included in the headers, then the custom range is used instead. If the file is small enough, then the entire file is sent as a 200.

Handling Multiple Clients

Our server handles multiple clients by launching threads for each connection. Each connection is handled by that thread until the connection is closed. We believe that this method is the most efficient and can easily handle thousands of clients.

Libraires

No external libraries were used.

Extra Capabilities

No extra capabilities were added.